

常微分方程讲义



常微分方程期末复习 (三)



嘉应学院数学学院

<https://math.jyu.edu.cn>

第五章 (30----36分)

一、存在唯一性定理

1 一阶线性微分方程组

$$x'(t) = A(t)x(t) + f(t)$$

(1) 解的存在唯一性定理

$$\begin{cases} x'(t) = A(t)x(t) + f(t) \\ x(t_0) = \eta \end{cases}$$

(2) 等价的积分方程

$$x(t) = \eta + \int_{t_0}^t (A(s)x(s) + f(s))ds$$



2 n阶线性微分方程的与一阶线性微分方程组的等价关系

$$x^{(n)} + a_1(t)x^{(n-1)} + \cdots + a_n(t)x = f(t) \quad (5.6)$$

$$\Downarrow x_1 = x, x_2 = x', x_3 = x'', \cdots, x_n = x^{(n-1)}.$$

$$X' = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n(t) & -a_{n-1}(t) & -a_{n-2}(t) & \cdots & -a_1(t) \end{bmatrix} X + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ f(t) \end{bmatrix} \quad (5.7)$$

$$\text{若 } x = \varphi(t) \text{ 为 (5.6) 的解} \Rightarrow X = \begin{bmatrix} \varphi(t) \\ \varphi'(t) \\ \vdots \\ \varphi^{(n-1)}(t) \end{bmatrix} \text{ 为 (5.7) 的解}$$

$$\text{若 } X = \begin{bmatrix} \varphi_1(t) \\ \varphi_2(t) \\ \vdots \\ \varphi_n(t) \end{bmatrix} \text{ 为 (5.7) 的解} \Rightarrow x = \varphi_1(t) \text{ 为 (5.6) 的解}$$



例: 将二阶微分方程 $x'' + 2x' + 7tx = e^{-t}$ 化为与之等价的微分方程组.

解: 令 $x_1 = x, x_2 = x'$

$$\text{则 } x_1' = x_2, x_2' = x'' = -7tx - 2x' + e^{-t} = -7tx_1 - 2x_2 + e^{-t}$$

即与之**等价的微分方程组**为:
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ -7t & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ e^{-t} \end{bmatrix}$$

注: 若 $X = \begin{bmatrix} \varphi_1(t) \\ \varphi_2(t) \end{bmatrix}$ 为方程组 $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ -7t & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ e^{-t} \end{bmatrix}$ 的一个解

则 $x = \varphi_1(t)$ 为方程 $x'' + 2x' + 7tx = e^{-t}$ 的解.

反: 若 $x = \varphi(t)$ 为方程 $x'' + 2x' + 7tx = e^{-t}$ 的解,

则 $X = \begin{bmatrix} \varphi(t) \\ \varphi'(t) \end{bmatrix}$ 为方程组 $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ -7t & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ e^{-t} \end{bmatrix}$ 的一个解



二、线性微分方程组的一般理论

(一) 齐线性微分方程组解的理论

$$x'(t) = A(t)x(t) \quad (5.15)$$

1 解的性质

2 (5.15)的 n 个解 $x_1(t), x_2(t) \cdots, x_n(t)$ 在 $[a, b]$ 上线性无关 $\Leftrightarrow$

$$W(t) = W[x_1(t), x_2(t) \cdots, x_n(t)] \neq 0, t \in [a, b]$$

设 $x_1(t), x_2(t) \cdots, x_n(t)$ 为(5.15)的 n 个解

则 $W'(t) = (a_{11}(t) + \cdots + a_{nn}(t))W(t)$ ($n=2$ 会证明)

$$W(t) = W(t_0)e^{\int_{t_0}^t (a_{11}(s) + \cdots + a_{nn}(s))ds}$$

3 通解 (1) 基本解组 (2) 通解结构

设 $x_1(t), x_2(t) \cdots, x_n(t)$ 为(5.15)的 n 个线性无关解, 则通解为

$$x(t) = c_1 x_1(t) + c_2 x_2(t) + \cdots + c_n x_n(t)$$



证明

因 $x_1(t) = \begin{bmatrix} x_{11}(t) \\ x_{21}(t) \end{bmatrix}$, $x_2(t) = \begin{bmatrix} x_{12}(t) \\ x_{22}(t) \end{bmatrix}$ 为方程 $x' = \begin{bmatrix} a_{11}(t) & a_{12}(t) \\ a_{21}(t) & a_{22}(t) \end{bmatrix} x$ 解

$$\text{故 } \begin{bmatrix} x'_{11}(t) \\ x'_{21}(t) \end{bmatrix} = \begin{bmatrix} a_{11}(t) & a_{12}(t) \\ a_{21}(t) & a_{22}(t) \end{bmatrix} \begin{bmatrix} x_{11}(t) \\ x_{21}(t) \end{bmatrix}, \begin{bmatrix} x'_{12}(t) \\ x'_{22}(t) \end{bmatrix} = \begin{bmatrix} a_{11}(t) & a_{12}(t) \\ a_{21}(t) & a_{22}(t) \end{bmatrix} \begin{bmatrix} x_{12}(t) \\ x_{22}(t) \end{bmatrix}$$

$$\text{从而 } W'(t) = \begin{vmatrix} x_{11}(t) & x_{12}(t) \\ x_{21}(t) & x_{22}(t) \end{vmatrix}' = \begin{vmatrix} x'_{11}(t) & x'_{12}(t) \\ x_{21}(t) & x_{22}(t) \end{vmatrix} + \begin{vmatrix} x_{11}(t) & x_{12}(t) \\ x'_{21}(t) & x'_{22}(t) \end{vmatrix}$$

$$\begin{aligned} \text{从而 } W'(t) &= \begin{vmatrix} a_{11}(t)x_{11}(t) + a_{12}(t)x_{21}(t) & a_{11}(t)x_{12}(t) + a_{12}(t)x_{22}(t) \\ x_{21}(t) & x_{22}(t) \end{vmatrix} \\ &\quad + \begin{vmatrix} x_{11}(t) & x_{12}(t) \\ a_{21}(t)x_{11}(t) + a_{22}(t)x_{21}(t) & a_{21}(t)x_{12}(t) + a_{22}(t)x_{22}(t) \end{vmatrix} \end{aligned}$$



$$= \begin{vmatrix} a_{11}(t)x_{11}(t) & a_{11}(t)x_{12}(t) \\ x_{21}(t) & x_{22}(t) \end{vmatrix} + \begin{vmatrix} x_{11}(t) & x_{12}(t) \\ a_{22}(t)x_{21}(t) & a_{22}(t)x_{22}(t) \end{vmatrix}$$

$$= a_{11}(t) \begin{vmatrix} x_{11}(t) & x_{12}(t) \\ x_{21}(t) & x_{22}(t) \end{vmatrix} + a_{22}(t) \begin{vmatrix} x_{11}(t) & x_{12}(t) \\ x_{21}(t) & x_{22}(t) \end{vmatrix}$$

$$= [a_{11}(t) + a_{22}(t)]W(t)$$

$$\text{由 } W'(t) = [a_{11}(t) + a_{22}(t)]W(t) \text{ 得 } \frac{1}{W(t)} \frac{dW(t)}{dt} = a_{11}(t) + a_{22}(t)$$

$$\text{两边从 } t_0, t \text{ 积分得 } \ln W(t) - \ln W(t_0) = \int_{t_0}^t [a_{11}(s) + a_{22}(s)] ds$$

$$\text{故: } W(t) = W(t_0) e^{\int_{t_0}^t [a_{11}(s) + a_{22}(s)] ds}$$



4 基解矩阵 $\Phi(t) = [x_1(t), x_2(t) \cdots, x_n(t)]$.

(1) $\Phi(t)$ 为 (5.15) 的基解矩阵 $\Leftrightarrow \begin{cases} \Phi'(t) = A(t)\Phi(t) \\ \det \Phi(t) \neq 0 \end{cases}$

(2) 通解 $x = \Phi(t)C$,

$$\varphi(t_0) = \eta \text{ 的解, } \varphi(t) = \Phi(t)\Phi^{-1}(t_0)\eta$$

(3) 性质

1⁰ 设 $\Phi(t)$ 为 (5.15) 的基解矩阵, C 是非奇异的, 则 $\Phi(t)C$ 也是 (5.15) 的基解矩阵.

2⁰ 若 $\Phi(t), \Psi(t)$ 是基解矩阵, 则 $\Psi(t) = \Phi(t)C$, 即 $[\Phi^{-1}(t)\Psi(t)]' = 0$ (会证)



证明 因 $\Phi(t), \Psi(t)$ 是基解矩阵, 故可设 $\Phi^{-1}(t)\Psi(t) = C(t)$

即 $\Psi(t) = \Phi(t)C(t)$, 代入方程 $x' = A(t)x$ 得

$$\Psi'(t) = \Phi'(t)C(t) + \Phi(t)C'(t) = A(t)\Phi(t)C(t) + \Phi(t)C'(t)$$

$$\Psi'(t) = A(t)\Psi(t) = A(t)\Phi(t)C(t)$$

从而 $\Phi(t)C'(t) = 0$

故 $C'(t) = 0$ 从而 $C(t) = C$

故 $[\Phi^{-1}(t)\Psi(t)]' = 0, \Psi(t) = \Phi(t)C.$



例: 设矩阵 $\Phi(t)$ 为方程组 $\frac{dx}{dt} = A(t)x$ 的一个基解矩阵,
则 $A(t)$ 由 $\Phi(t)$ 唯一确定.

证明: 由于 $\Phi(t)$ 为方程组 $\frac{dx}{dt} = A(t)x$ 的一个基解矩阵,

$$\text{故 } \Phi'(t) = A(t)\Phi(t),$$

且 $\det \Phi(t) \neq 0$, 即 $\Phi(t)$ 可逆,

$$\text{从而 } A(t) = \Phi'(t)\Phi^{-1}(t),$$

即 $A(t)$ 由 $\Phi(t)$ 唯一确定.



(二) 非齐线性微分方程组解的理论

$$x'(t) = A(t)x(t) + f(t) \quad (5.14)$$

1 解的性质

$$x'(t) = A(t)x(t) \quad (5.15)$$

2 通解结构

设 $\Phi(t)$ 为(5.15)的基解矩阵, $\bar{\varphi}(t)$ 是(5.14)的解,则通解

$$x(t) = \Phi(t)C + \bar{\varphi}(t)$$

3 常数变易公式

(5.14)满足 $\varphi(t_0) = 0$ 的解为

$$\varphi(t) = \Phi(t) \int_{t_0}^t \Phi^{-1}(s) f(s) ds \quad (\text{会证})$$

(5.14)满足 $\varphi(t_0) = \eta$ 的解为

$$\varphi(t) = \Phi(t)\Phi^{-1}(t_0)\eta + \Phi(t) \int_{t_0}^t \Phi^{-1}(s) f(s) ds \quad (\text{会证})$$



证明 设 $\varphi(t) = \Phi(t)c(t)$ 为(5.14)的解, 代入方程得

$$\Phi'(t)c(t) + \Phi(t)c'(t) = A(t)\Phi(t)c(t) + f(t)$$

$$\text{因 } \Phi'(t) = A(t)\Phi(t), \quad \text{故 } c'(t) = \Phi^{-1}(t)f(t)$$

$$\text{从而: } c(t) = c(t_0) + \int_{t_0}^t \Phi^{-1}(s)f(s)ds$$

$$\text{故: } \varphi(t) = \Phi(t)c(t_0) + \int_{t_0}^t \Phi(t)\Phi^{-1}(s)f(s)ds$$

$$\text{由 } \varphi(t_0) = \eta, \quad \text{得 } c(t_0) = \Phi^{-1}(t_0)\eta,$$

$$\text{故 } \varphi(t) = \Phi(t)\Phi^{-1}(t_0)\eta + \int_{t_0}^t \Phi(t)\Phi^{-1}(s)f(s)ds$$

注: 由 $\varphi(t_0) = 0$, 得 $c(t_0) = 0$,

$$\text{故 } \varphi(t) = \int_{t_0}^t \Phi(t)\Phi^{-1}(s)f(s)ds$$



4 二阶线性微分方程的常数变易公式

设 $x_1(t), x_2(t)$ 为 $x'' + a_1(t)x' + a_2(t)x = f(t)$ 对应齐次方程的基本解组, 则满足 $\varphi(t_0) = 0, \varphi'(t_0) = 0$ 的解为。

$$\varphi(t) = \int_{t_0}^t \frac{x_2(t)x_1(s) - x_1(t)x_2(s)}{W[x_1(s), x_2(s)]} f(s) ds, \quad (5.31)$$

三、常系数线性方程组

$$\frac{dx}{dt} = Ax, \quad (5.33)$$

(一) 基解矩阵

$$\exp At = \sum_{k=0}^{\infty} \frac{A^k}{k!} t^k = E + At + \frac{A^2}{2!} t^2 + \cdots + \frac{A^m}{m!} t^m + \cdots \quad (5.34)$$

2 通解为 $x = (\exp At)C$

$\varphi(t_0) = \eta$ 的解为 $\varphi(t) = \exp A(t - t_0)\eta$

3 $\exp At$ 性质

(1) 若 $AB = BA$, 则 $\exp(A + B) = \exp A \exp B$.

(2) 若 A 为对角矩阵, 则 $\exp A =$,

(3) 若 $\Phi(t)$ 为 (5.33) 的基解矩阵, 则 $\exp At = \Phi(t)\Phi^{-1}(0)$



(二) 基解矩阵的计算

$\varphi(t) = e^{\lambda t} c \neq 0$ 为(5.33)解 $\Leftrightarrow (\lambda E - A)c = 0$ 有非零解,

λ 为 A 的特征值, c 为相应的特征向量,

1 若矩阵 A 有 n 个线性无关的特征向量 $v_1, v_2, \dots, v_n$,
相应的特征值为 $\lambda_1, \lambda_2, \dots, \lambda_n$, 则其基解矩阵为

$$\Phi(t) = [e^{\lambda_1 t} v_1, e^{\lambda_2 t} v_2, \dots, e^{\lambda_n t} v_n],$$

$\exp At = \Phi(t)\Phi^{-1}(0)$, (基解矩阵的标准(实数)化)

注 若矩阵 A 有 n 个互不相同特征值 $\lambda_1, \lambda_2, \dots, \lambda_n$, 相应的特征向量为 $v_1, v_2, \dots, v_n$, 则其基解矩阵为

$$\Phi(t) = [e^{\lambda_1 t} v_1, e^{\lambda_2 t} v_2, \dots, e^{\lambda_n t} v_n],$$



2 若 A 的特征值为 $\lambda_1, \lambda_2, \dots, \lambda_k$, 相应的重数为 $n_1, n_2, \dots, n_k$,

$$\text{令: } U_j = \{u \in U \mid (A - \lambda_j E)^{n_j} u = 0\}$$

(5.33)满足 $\varphi(0) = \eta$ 的解为

$$\text{设: } \eta = v_1 + v_2 + \dots + v_k, v_j \in U_j$$

$$\varphi(t) = (\exp At)\eta = \sum_{j=1}^k e^{\lambda_j t} \left\{ \sum_{l=0}^{n_j-1} \frac{t^l}{l!} (A - \lambda_j E)^l \right\} v_j$$

$$\exp At = [(\exp At)e_1, (\exp At)e_2, \dots, (\exp At)e_n]$$

3 若 A 只有一个 n 重特征值 λ , 则

$$\exp At = e^{\lambda t} \sum_{k=0}^{n-1} \frac{t^k}{k!} (A - \lambda E)^k$$



4 $n = 2$ 的情形

(1) 若 A 有两个互不相同特征值为 λ_1, λ_2 , 相应的特征向量为 v_1, v_2 , 则其基解矩阵为

$$\Phi(t) = [e^{\lambda_1 t} v_1, e^{\lambda_2 t} v_2],$$

$$\text{且: } \exp At = \Phi(t)\Phi^{-1}(0),$$

(2) 若 A 只有一个二重特征值 λ , 则

$$\exp At = e^{\lambda t} [E + (A - \lambda E)t]$$

例12 求微分方程组 $x' = \begin{bmatrix} 4 & -2 \\ 2 & -1 \end{bmatrix} x$ 的基解矩阵 $\exp At$.

解 A 的特征方程为: $\det(\lambda E - A) = \begin{vmatrix} \lambda - 4 & 2 \\ -2 & \lambda + 1 \end{vmatrix} = \lambda^2 - 3\lambda = 0$

得特征值 $\lambda_1 = 0, \lambda_2 = 3$,

对特征值 λ_1, λ_2 , 相应特征向量 $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ 满足

$$(\lambda_1 E - A)u = \begin{bmatrix} -4 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad (\lambda_2 E - A)v = \begin{bmatrix} -1 & 2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

解得 $u = \alpha \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad v = \beta \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \alpha, \beta$ 为不为零的任常数



故微分方程组 $x' = Ax$ 的基解矩阵为

$$\Phi(t) = \begin{bmatrix} 1 & 2e^{3t} \\ 2 & e^{3t} \end{bmatrix}$$

$$\begin{aligned} \text{从而: } \exp At &= \Phi(t)\Phi^{-1}(0) = \begin{bmatrix} 1 & 2e^{3t} \\ 2 & e^{3t} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}^{-1} \\ &= -\frac{1}{3} \begin{bmatrix} 1 & 2e^{3t} \\ 2 & e^{3t} \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \\ &= -\frac{1}{3} \begin{bmatrix} 1-4e^{3t} & -2+2e^{3t} \\ 2-2e^{3t} & -4+e^{3t} \end{bmatrix} \end{aligned}$$

(三) 非齐

$$x' = Ax + f(t), \quad (5.60)$$

满足初始条件 $\varphi(t_0) = \eta$ 的解

$$\varphi(t) = \exp A(t-t_0)\eta + \int_{t_0}^t \exp A(t-s)f(s)ds$$

例13: 设 $A = \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}$, $f(t) = \begin{bmatrix} 2e^t \\ e^t \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(1) 求方程组 $x' = Ax$ 的基解矩阵 $\exp At$,

(2) 求初值问题 $\begin{cases} x' = Ax + f(t) \\ x(0) = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \end{cases}$ 的解.



解:

(1) 矩阵 A 的特征方程为: $\det(\lambda E - A) = \begin{vmatrix} \lambda - 1 & 2 \\ 1 & \lambda - 2 \end{vmatrix} = \lambda^2 - 3\lambda = 0$

得特征值 $\lambda_1 = 0, \lambda_2 = 3$, 相应的特征向量满足

$$(\lambda_1 E - A)u = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad (\lambda_2 E - A)v = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

解之得: $u = \alpha \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad v = \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \alpha, \beta$ 为不等于零的任常数,

故方程组 $x' = Ax$ 的基解矩阵为: $\Phi(t) = \begin{bmatrix} 2 & e^{3t} \\ 1 & -e^{3t} \end{bmatrix}$

$$\text{从而 } \exp At = \Phi(t)\Phi^{-1}(0) = -\frac{1}{3} \begin{bmatrix} 2 & e^{3t} \\ 1 & -e^{3t} \end{bmatrix} \begin{bmatrix} -1 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 + e^{3t} & 2 - 2e^{3t} \\ 1 - e^{3t} & 1 + 2e^{3t} \end{bmatrix}$$



(2)由常数变易公式

$$x(t) = \exp[A(t-t_0)]\eta + \int_{t_0}^t \exp[A(t-s)]f(s)ds$$

得初值问题的解为

$$\begin{aligned} x(t) &= \exp(At)\eta + \int_0^t \exp[A(t-s)]f(s)ds \\ &= \frac{1}{3} \begin{bmatrix} 2+e^{3t} & 2-2e^{3t} \\ 1-e^{3t} & 1+2e^{3t} \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{1}{3} \int_0^t \begin{bmatrix} 2+e^{3(t-s)} & 2-2e^{3(t-s)} \\ 1-e^{3(t-s)} & 1+2e^{3(t-s)} \end{bmatrix} \begin{bmatrix} 2e^s \\ e^s \end{bmatrix} ds \\ &= \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \int_0^t \begin{bmatrix} 2e^s \\ e^s \end{bmatrix} ds = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 2e^t - 2 \\ e^t - 1 \end{bmatrix} = \begin{bmatrix} 2e^t \\ e^t \end{bmatrix} \end{aligned}$$



例14: 设 $A = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$, $f(t) = \begin{bmatrix} e^{3t} \\ e^{3t} \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(1) 求方程组 $x' = Ax$ 的基解矩阵 $\exp At$,

(2) 求初值问题 $\begin{cases} x' = Ax + f(t) \\ x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{cases}$ 的解.

解:

(1) 矩阵 A 的特征方程为: $\det(\lambda E - A) = \begin{vmatrix} \lambda - 2 & -1 \\ 1 & \lambda - 4 \end{vmatrix} = \lambda^2 - 6\lambda + 9 = 0$

得二重特征值为 $\lambda = 3$, 故其实基解矩阵 $\exp At$ 为

$$\begin{aligned} \exp At &= e^{3t} [E + t(A - 3E)] \\ &= e^{3t} \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + t \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \right\} = e^{3t} \begin{bmatrix} 1-t & t \\ -t & 1+t \end{bmatrix} \end{aligned}$$



(2)由常数变易公式

$$x(t) = \exp[A(t-t_0)]\eta + \int_{t_0}^t \exp[A(t-s)]f(s)ds$$

得初值问题的解为

$$\begin{aligned} x(t) &= \exp(At)\eta + \int_0^t \exp[A(t-s)]f(s)ds \\ &= e^{3t} \begin{bmatrix} 1-t & t \\ -t & 1+t \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \int_0^t e^{3(t-s)} \begin{bmatrix} 1-(t-s) & t-s \\ -(t-s) & 1+(t-s) \end{bmatrix} \begin{bmatrix} e^{3s} \\ e^{3s} \end{bmatrix} ds \\ &= e^{3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \int_0^t \begin{bmatrix} e^{3t} \\ e^{3t} \end{bmatrix} ds = e^{3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^{3t} \begin{bmatrix} t \\ t \end{bmatrix} = e^{3t} \begin{bmatrix} 1+t \\ 1+t \end{bmatrix} \end{aligned}$$

练习:P245 7,11



谢谢大家！

